【解答】対数の定義　指数 ⇄ 対数

数学 II ｜ 対数関数 ①

組　番　名前

1　log を使った式で表す

① $2^3 = 8 \Rightarrow \log_2 8 = 3$

② $3^4 = 81 \Rightarrow \log_3 81 = 4$

③ $5^2 = 25 \Rightarrow \log_5 25 = 2$

④ $10^3 = 1000 \Rightarrow \log_{10} 1000 = 3$

⑤ $7^0 = 1 \Rightarrow \log_7 1 = 0$

⑥ $4^1 = 4 \Rightarrow \log_4 4 = 1$

⑦ $2^{-3} = \frac{1}{8} \Rightarrow \log_2 \frac{1}{8} = -3$

⑧ $3^{-2} = \frac{1}{9} \Rightarrow \log_3 \frac{1}{9} = -2$

⑨ $9^{\frac{1}{2}} = 3 \Rightarrow \log_9 3 = \frac{1}{2}$

⑩ $8^{\frac{1}{3}} = 2 \Rightarrow \log_8 2 = \frac{1}{3}$

⑪ $2^{\frac{1}{2}} = \sqrt{2} \Rightarrow \log_2 \sqrt{2} = \frac{1}{2}$

⑫ $(\frac{1}{2})^3 = \frac{1}{8} \Rightarrow \log_{(\frac{1}{2})} \frac{1}{8} = 3$

2　$a^p = M$ の形で表す

① $\log_2 32 = 5 \Rightarrow 2^5 = 32$

② $\log_3 27 = 3 \Rightarrow 3^3 = 27$

③ $\log_{10} 100 = 2 \Rightarrow 10^2 = 100$

④ $\log_5 1 = 0 \Rightarrow 5^0 = 1$

⑤ $\log_6 \frac{1}{36} = -2 \Rightarrow 6^{-2} = \frac{1}{36}$

⑥ $\log_4 8 = \frac{3}{2} \Rightarrow 4^{\frac{3}{2}} = 8$

⑦ $\log_{(\frac{1}{3})} 9 = -2 \Rightarrow (\frac{1}{3})^{-2} = 9$

⑧ $\log_2 \frac{1}{\sqrt{2}} = -\frac{1}{2} \Rightarrow 2^{-\frac{1}{2}} = \frac{1}{\sqrt{2}}$

【解答】対数の値を求める

数学 II ｜ 対数関数 ②

組　番　名前

3　次の値を求めなさい

① $\log_2 16 \Rightarrow \mathbf{4}$

② $\log_3 9 \Rightarrow \mathbf{2}$

③ $\log_5 125 \Rightarrow \mathbf{3}$

④ $\log_{10} 10000 \Rightarrow \mathbf{4}$

⑤ $\log_7 7 \Rightarrow \mathbf{1}$

⑥ $\log_6 1 \Rightarrow \mathbf{0}$

⑦ $\log_2 \frac{1}{4} \Rightarrow \mathbf{-2}$

⑧ $\log_3 \frac{1}{27} \Rightarrow \mathbf{-3}$

⑨ $\log_4 2 \Rightarrow \mathbf{\frac{1}{2}}$

⑩ $\log_9 27 \Rightarrow \mathbf{\frac{3}{2}}$

⑪ $\log_2 \sqrt{8} \Rightarrow \mathbf{\frac{3}{2}}$

⑫ $\log_{(\frac{1}{2})} 8 \Rightarrow \mathbf{-3}$

⑬ $\log_{(\frac{1}{3})} \frac{1}{81} \Rightarrow \mathbf{4}$

⑭ $\log_{25} 5 \Rightarrow \mathbf{\frac{1}{2}}$

⑮ $\log_8 4 \Rightarrow \mathbf{\frac{2}{3}}$

⑯ $\log_{\sqrt{2}} 8 \Rightarrow \mathbf{6}$

4　x の値を求めなさい

① $\log_2 x = 5 \Rightarrow \mathbf{x = 32}$

② $\log_3 x = -2 \Rightarrow \mathbf{x = \frac{1}{9}}$

③ $\log_x 49 = 2 \Rightarrow \mathbf{x = 7}$

④ $\log_x 8 = 3 \Rightarrow \mathbf{x = 2}$

⑤ $\log_5 x = 0 \Rightarrow \mathbf{x = 1}$

⑥ $\log_x \frac{1}{16} = -4 \Rightarrow \mathbf{x = 2}$

⑦ $\log_4 x = \frac{3}{2} \Rightarrow \mathbf{x = 8}$

⑧ $\log_x 27 = \frac{3}{2} \Rightarrow \mathbf{x = 9}$